Group-27: Livestock Monitoring System

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Contents

1 Overview **3**

2 Objectives **3**

3 Working Modules **3**

3.1 Collar Node 3

3.2 Edge Node 4

3.3 Gateway Node (Base Station) 4

3.4 Supporting Components 4

4 Software Required **4**

4.1 Gateway-Side Software 4

4.2 App and Development Tools 5

5 Libraries Used **5**

5.1 Edge Node Side 5

5.2 Collar Node Side 5

6 Mathematical Model: RSSI and Distance **5**

6.1 Experimental Calibration 5

6.2 Experimental Results Graph 6

7 Signal Processing Challenges **6**

7.1 Problems in RSSI Calculation 7

8 Advanced Signal Processing Pipeline **7**

8.1 Step 1: Median Filter and Boundary Preservation 7
8.2 Step 2: EWMA and Z-Transform Analysis 8
8.3 Step 3: Monte Carlo Simulation 9

9 Future Enhancements **9**

1 Overview

Herders often face significant challenges in managing large livestock herds. As the number of animals increases, it becomes difficult to continuously monitor each individual while they graze freely in open fields. Animals may wander outside the designated area, get lost, or even fall prey to predators before the herder notices anything is wrong.

To address this issue, this project aims to develop a **LoRa-based Livestock Monitoring System** that enables long-range tracking of each animal in the herd. Each animal will be fitted with a small collar device containing a LoRa transceiver and microcontroller. These devices send periodic location updates to a gateway system, which then forwards the data to a mobile application used by the herder.

Through the app, the herder can view the real-time location of every animal. If any animal moves beyond the defined grazing perimeter, the system immediately triggers an alert or notification. This greatly improves the herder's ability to supervise large herds efficiently, reduce losses, and ensure that the animals remain within safe boundaries.

2 Objectives

1. **Establish Wireless Communication:** LoRa chips are used to establish a long-range wireless connection between the livestock and the herder.
2. **Monitor Livestock:** Track the location of animals using the LoRa chips.
3. **Filter Noise:** The bandwidth that the LoRa chips use is vulnerable to noise. We are tasked with filtering the noise to give accurate measurements of distance.
4. **Develop a Dashboard:** We aim to set up a web-based dashboard/app to track distance and temperature

3 Working Modules

The system is built upon three core working modules: the **Collar Node**, the **Edge Node**, and the **Gateway Node**. Each module comprises specific internal parts essential for its operation.

3.1 Collar Node

Internal Part	Qty	Purpose
Arduino UNO	1	Microcontroller to drive the LoRa chips.
LoRa Module SX1278	1	Used to transmit packets to the edge node.
Antenna	1	Ensures correct long-range communication (865-867MHz rated).

3.2 Edge Node

Internal Part	Qty	Purpose
ESP32	1	Microcontroller with WiFi for forwarding the received LoRa data.
LoRa Module SX1278	1	Receives the LoRa packets from the collar.
Antenna	1	Ensures correct long-range communication.

3.3 Gateway Node (Base Station)

Internal Part	Qty	Purpose
Raspberry Pi 3	1	Acts as the gateway server. Runs a Python script to receive data over WiFi. Signal processing is done here.
Micro SD card	1	Contains the Raspberry Pi OS.

3.4 Supporting Components

- Jumper Wires for all connections.
- Power/Data Cables to connect all nodes with a stable power supply and to display the serial monitor.
- Soldering tools.
- Assembly tools.
- Cardboard, Fevicol, etc.

4 Software Required

4.1 Gateway-Side Software

Software / Tool	Purpose / Use
Arduino IDE	For programming ESP32 and Arduino-based collars and edge server.
Raspberry Pi OS	Operating system for the Raspberry Pi gateway.
Python (Flask / FastAPI)	For building local APIs to handle data from LoRa modules.
Firebase / MQTT Broker	For real-time data sync between gateway and mobile app.

4.2 App and Development Tools

Software / Tool	Purpose / Use
Android Studio	To build and test the farmer's mobile app (Android).
Figma	For designing app UI/UX.
Firebase Realtime Database	For storing and syncing sensor data in the cloud.
Postman	To test and debug API endpoints from the gateway.

5 Libraries Used

5.1 Edge Node Side

Purpose	Library Name
LoRa reception	`LoRa.h`
Wireless to phone (WiFi)	`WiFi.h`
Create HTTP server	`WebServer.h` / `ESPAsyncWebServer.h`
Bluetooth alternative	`BluetoothSerial.h`
JSON formatting for app packets	`ArduinoJson.h`

5.2 Collar Node Side

Purpose	Library Name
LoRa communication	`LoRa.h`
GPS parsing	`TinyGPS++`
Serial communication for GPS	`SoftwareSerial`

6 Mathematical Model: RSSI and Distance

The system utilizes Received Signal Strength Indicator (RSSI) values to estimate the distance between the collar and the edge node. The relationship is governed by the Log-Distance Path Loss model.

6.1 Experimental Calibration

To ensure accurate distance estimation, we performed calibration experiments under controlled room conditions. RSSI readings were recorded at fixed distances (15cm, 30cm, 45cm, and 60cm) to determine the environmental parameters.

Using the Least Square Approximation method on the average RSSI readings, we derived the parameters. The calculation was performed using the Normal Equation:

$$A^T Ax = A^T y \tag{1}$$

where A is the design matrix, y is the vector of observed RSSI values, and x contains the parameters (n and A) to be estimated.

The derived parameters are:

- **Path Loss Exponent (n):** 1.72
- **Reference RSSI at 1 meter (A):** -83.82 dBm

The calibrated equation governing the relationship is:

$$RSSI = -10n \log_{10}(d) + A \tag{2}$$

Substituting our experimental values:

$$RSSI = -17.2 \log_{10}(d) - 83.82 \tag{3}$$

Where d is the distance in meters.

6.2 Experimental Results Graph

The following graph illustrates the experimental data points collected at fixed intervals and the fitted curve derived from the calculated parameters.

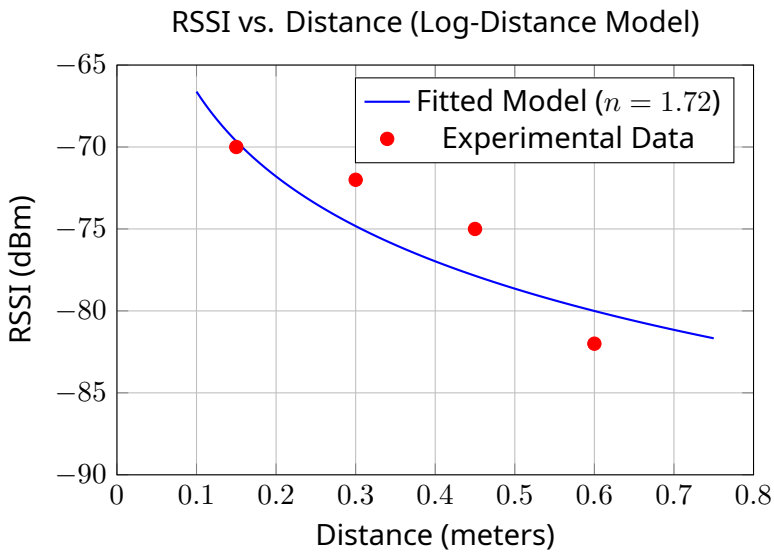

Figure 1: Graph showing the experimental RSSI readings (red dots) and the Least Square Approximation curve (blue line).

Rearranging the formula to solve for distance (d) based on real-time RSSI:

$$d = 10^{\left(\frac{A - RSSI}{10n}\right)} = 10^{\left(\frac{-83.82 - RSSI}{17.2}\right)} \tag{4}$$

7 Signal Processing Challenges

While the Log-Distance model provides a theoretical framework for distance estimation, real-world RSSI readings are inherently unstable. This section outlines the significant challenges faced during RSSI calculation which make raw data unreliable for precise distance estimation.

7.1 Problems in RSSI Calculation

1. **Multipath Interference**
   - **Phenomenon:** Signals bounce off obstacles (walls, ground, animals), creating multiple delayed copies of the same signal at the receiver.
   - **Impact:** These copies can add up or cancel each other out (constructive/destructive interference), causing the RSSI to fluctuate heavily even when the device is stationary.
2. **Fading (Shadowing & Fast Fading)**
   - **Slow Fading (Shadowing):** Caused by large obstacles (e.g., a cow's body or a tree) blocking the line of sight. This causes RSSI to drop for seconds or minutes, leading to overestimated distances.
   - **Fast Fading:** Caused by small-scale movements (1–5 cm). This results in rapid millisecond-level variations, making the signal look "jittery."
3. **Environmental Dependence**
   - RSSI is sensitive to humidity, temperature, and antenna orientation. A distance that reads -70 dBm on a dry day might read -75 dBm on a humid day.
4. **Non-linearity of Path Loss**
   - The path loss exponent n is not constant in dynamic environments. The mapping from RSSI to distance is non-linear, meaning small RSSI errors at long ranges result in massive distance errors.

8 Advanced Signal Processing Pipeline

While a Kalman Filter was initially implemented to address the noise issues described above, experimental trials revealed significant limitations. The filter struggled with the non-Gaussian nature of the noise (caused by multipath interference) and the high non-linearity of the RSSI-distance relationship. It often failed to react quickly to sudden movements or, conversely, over-reacted to spurious noise spikes.

To effectively solve these problems, we replaced the Kalman Filter with a robust three-stage pipeline: **Median Filter** $\rightarrow$ **EWMA** $\rightarrow$ **Monte Carlo Simulation**.

8.1 Step 1: Median Filter and Boundary Preservation

The first stage involves passing the raw RSSI data through a Median Filter.

Boundary Point Preservation: Unlike Least Squares or Mean filters, which calculate averages, the Median Filter sorts values in a window and picks the middle one. This makes it exceptionally good at handling *boundary points*—sudden, sharp changes in the signal (e.g., if the animal suddenly sprints).

- **Least Squares/Mean:** Would average the new "boundary" value with previous values, creating a "sloped" or blurred transition. This introduces lag and inaccuracies during rapid movement.

- **Median Filter:** Can accept the new value immediately once it dominates the window, preserving the sharp "edge" of the movement without the blurring effect. It effectively rejects impulsive noise (outliers) without smearing the signal boundaries.

8.2 Step 2: EWMA and Z-Transform Analysis

After removing outliers, the signal is smoothed using an Exponential Weighted Moving Average (EWMA) to remove high-frequency "jiggle."

The time-domain difference equation for EWMA is:

$$y[n] = \alpha x[n] + (1 - \alpha)y[n - 1] \tag{5}$$

Where $y[n]$ is the smoothed output, $x[n]$ is the input, and α is the smoothing factor ($0 < \alpha < 1$).

To understand how this removes "jiggle," we apply the **Z-transform**:

$$Y(z) = \alpha X(z) + (1 - \alpha)z^{-1}Y(z) \tag{6}$$

Rearranging to find the Transfer Function $H(z)$:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\alpha}{1 - (1 - \alpha)z^{-1}} = \frac{\alpha z}{z - (1 - \alpha)} \tag{7}$$

This transfer function represents a single-pole low-pass filter with a pole at $z = 1 - \alpha$.

- **Low Frequencies (DC):** At $z = 1$ (frequency 0), $|H(z)| \approx 1$. The signal passes through.
- **High Frequencies (Jiggle):** At $z = -1$ the magnitude drops significantly.

This proves mathematically that the filter attenuates rapid fluctuations (noise) while allowing the underlying trend (animal movement) to pass.

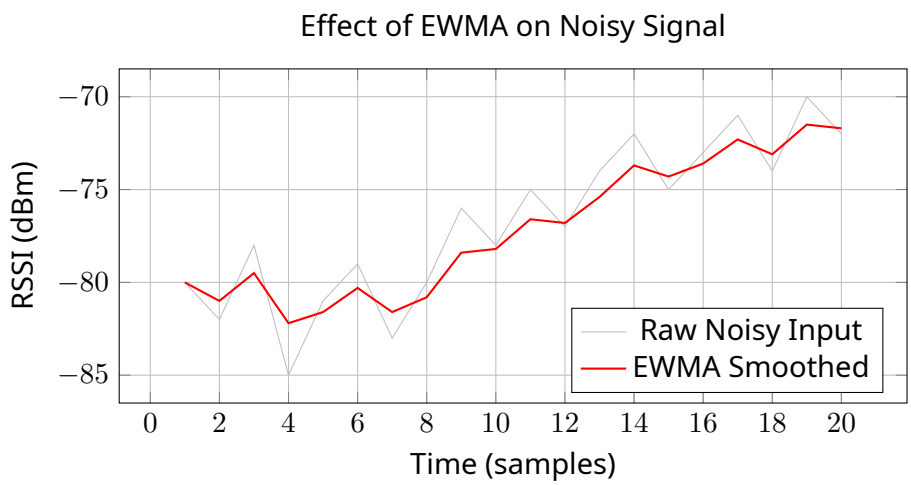

Figure 2: Comparison of Raw Signal "Jiggle" vs. EWMA Smoothed Output.

8.3 Step 3: Monte Carlo Simulation

The final stage utilizes a Monte Carlo simulation. Direct algebraic inversion of the path loss formula ($d = 10^{\frac{A-RSSI}{10n}}$) is problematic because the function is highly non-linear; a small symmetric noise in RSSI results in asymmetric, skewed errors in distance.

The Process: Instead of calculating one single distance, we assume the smoothed RSSI follows a probability distribution (Gaussian) defined by the EWMA mean and the measured variance.

1. We generate $N = 1000$ random RSSI samples based on the current smoothed value ($RSSI_{smooth}$) and noise deviation (σ).
2. We convert every single sample into a distance using the path loss model.
3. This creates a distribution of 1000 possible distances.
4. We analyze this distribution to find the most probable distance (the mode or median of the distribution).

This method effectively "simulates" the uncertainty and gives us a confidence interval rather than a single, potentially erroneous point.

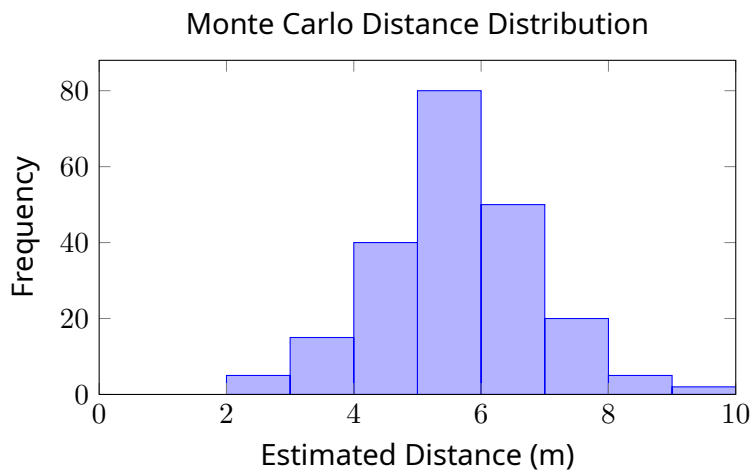

Figure 3: Histogram of potential distances generated by Monte Carlo Simulation for a single reading.

9 Future Enhancements

- **Additional Health Sensors:** Integrate sensors such as humidity or heart-rate monitors to gather more comprehensive health data, allowing early detection of stress or illness in livestock.
- **Solar-Powered Operation:** Equip the cattle devices with small solar panels to enable continuous, low-maintenance operation without frequent recharging.
- **Machine Learning-Based Behavior Analysis:** Apply machine learning models to analyze historical temperature and movement data, helping predict unusual behavior patterns or potential health issues before they become critical.